

Additional Informations

1 Further explanation of the set C of critical points in the network

The set C is defined to simplify the modeling of constraints that prevent collisions between vehicles. It contains the critical points in the network G' where collisions can occur and is composed of triples (k, q, v) , with $C \subseteq V \times V \times V$. Figure 1 illustrates an element $(k, q, v) \in C$ on a network with six stopping points, where $P = \{p_1, p_2, p_3, p_4, p_5, p_6\}$. Let $p_1, p_6 \in D$ denote depots, $p_2, p_5 \in X$ crossings, and $p_3, p_4 \in S$ stations. Therefore, p_2 and p_5 are adjacent crossings.

Let v and q be the vertices that belong to crossing p_2 , where v belongs to the upper section and q to the lower section. Let k be the vertex in the lower section of crossing p_5 . If a vehicle departs from p_2 to the right, i.e., from v , while another vehicle departs from p_5 to the left, i.e., from k , then the segments between crossings p_2 and p_5 become a potential collision area. To avoid such conflict, the vehicle at v can only depart after the other vehicle has reached q , assuming that the vehicle in k was the first to depart. Hence, (k, q, v) represents critical elements in a part of the network where collisions can occur.

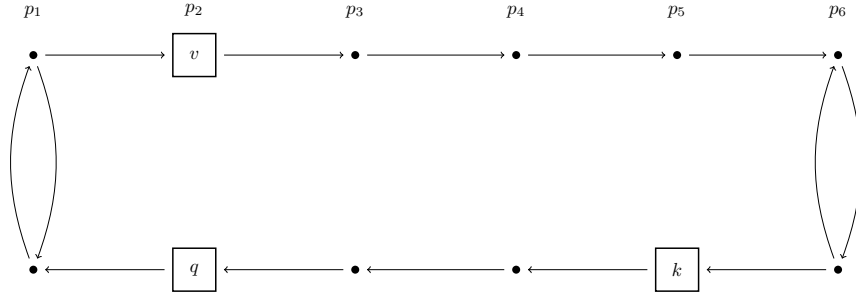


Figure 1: Triple $(k, q, v) \in C$, with v and q at crossing p_2 and k at crossing p_5 .

Figure 2 illustrates the same network, but now with (k, q, v) defined such that k belongs to crossing p_2 and v and q to crossing p_5 . In this case, the vertices and arcs that previously belonged to the upper section are now located in the lower section. This example shows that the vertices introduced in the previous explanation may belong either to the upper or in the lower section of the graph.

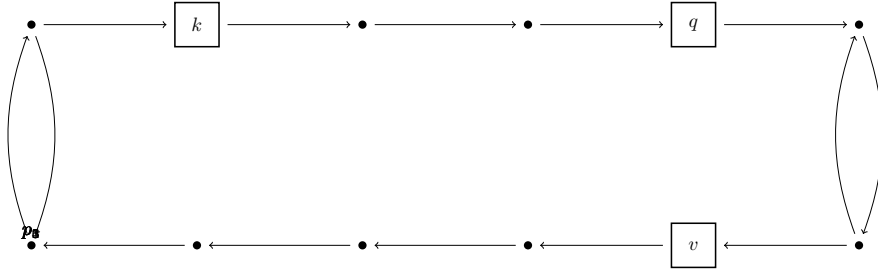


Figure 2: Triple $(k, q, v) \in C$, with v and q at crossing p_5 and k at crossing p_2 .

2 Detailed tables of results

Table 1: Results 5 to 9 stopping points with Gurobi.

Instance	Model				Enumeration of routes				MILP-based heuristic			
	UB	Gap	Gap*	Time	UB	Gap	Gap*	Time	UB	Gap	Gap*	Time
t2p5r2h1	31922	0.00%	0.00%	3.23	31922	0.00%	0.00%	1.40	31922	0.00%	0.00%	2.15
t2p6r7h3	42078	0.00%	0.00%	5.10	42078	0.00%	0.00%	3.79	42078	0.00%	0.00%	1.21
t2p7r8h1	17570	0.00%	0.00%	11.11	17570	0.00%	0.00%	161.70	17570	0.00%	0.00%	2.18
t2p8r2h2	20078	0.00%	0.00%	2.28	20078	0.00%	0.00%	1.37	20078	0.00%	0.00%	1.20
t2p9r2h3	37235	0.00%	0.00%	28.22	37235	0.00%	0.00%	4.42	37235	0.00%	0.00%	3.30
t3p5r7h1	7731	0.00%	0.00%	25.90	7731	0.00%	0.00%	16.06	7962	2.90%	2.90%	4.85
t3p6r2h1	23268	2.39%	0.00%	21601.08	23268	2.39%	0.00%	3958.16	23532	3.49%	1.12%	3604.09
t3p7r6h3	31411	0.00%	0.00%	25.85	31411	0.00%	0.00%	4.96	31411	0.00%	0.00%	3.55
t3p8r7h1	8474	0.00%	0.00%	13.12	8474	0.00%	0.00%	71.89	8474	0.00%	0.00%	5.26
t3p9r7h3	19386	0.00%	0.00%	4.22	19386	0.00%	0.00%	4.70	19386	0.00%	0.00%	2.36
t4p5r7h2	24488	10.58%	0.00%	21601.35	24488	10.58%	0.00%	21600.00	24488	10.58%	0.00%	3603.81
t4p6r7h1	31305	3.33%	0.00%	21602.72	31305	3.33%	0.00%	21600.01	31701	4.54%	1.25%	3607.48
t4p7r7h4	36082	0.00%	0.00%	379.75	36082	0.00%	0.00%	704.83	36082	0.00%	0.00%	7.63
t4p8r2h1	17313	0.01%	0.00%	8413.30	17313	0.01%	0.00%	4084.86	18001	3.83%	3.82%	3603.15
t4p9r2h1	9037	0.00%	0.00%	681.32	9037	0.00%	0.00%	176.80	9037	0.00%	0.00%	193.39
t5p5r2h5	37452	0.45%	0.00%	21600.88	37452	0.45%	0.00%	3606.76	37452	0.45%	0.00%	3601.84
t5p6r6h1	16818	17.17%	0.57%	21601.86	16722	16.69%	0.00%	23179.23	16809	17.13%	0.52%	3674.45
t5p7r2h2	22422	23.63%	4.25%	21602.06	21470	20.24%	0.00%	3614.38	22319	23.28%	3.80%	3610.33
t5p8r8h3	17469	0.00%	0.00%	3907.49	17469	0.00%	0.00%	4169.70	17469	0.00%	0.00%	3323.08
t5p9r2h3	15935	0.00%	0.00%	7031.76	15935	0.00%	0.00%	3607.28	15935	0.00%	0.00%	3602.84
Average	23374	2.88%	0.24%	7507.13	23321	2.68%	0.00%	4528.61	23447	3.31%	0.67%	1622.91

Table 2: Results 10 to 14 stopping points with Gurobi.

Instance	Model				Enumeration of routes				MILP-based heuristic			
	UB	Gap	Gap*	Time	UB	Gap	Gap*	Time	UB	Gap	Gap*	Time
t3p10r6h3	37612	0.00%	0.00%	43.52	37612	0.00%	0.00%	8.23	37612	0.00%	0.00%	4.24
t3p11r2h1	38040	0.00%	0.00%	8404.88	38040	0.00%	0.00%	3607.07	38040	0.00%	0.00%	1695.79
t3p12r2h3	36985	0.00%	0.00%	2.98	36985	0.00%	0.00%	3.43	36985	0.00%	0.00%	3.52
t3p13r12h2	17968	0.00%	0.00%	311.94	17968	0.00%	0.00%	72.09	17968	0.00%	0.00%	42.73
t3p14r8h5	31596	0.00%	0.00%	653.85	31596	0.00%	0.00%	13809.30	31596	0.00%	0.00%	67.41
t4p10r6h3	27620	0.00%	0.00%	3335.93	27620	0.00%	0.00%	173.12	27620	0.00%	0.00%	28.85
t4p11r2h2	40927	20.17%	0.00%	21601.80	40927	20.17%	0.00%	3611.27	40927	20.17%	0.00%	3608.00
t4p12r2h2	34230	0.01%	0.00%	18425.70	34230	0.01%	0.00%	3606.69	34230	0.01%	0.00%	3604.89
t4p13r8h1	31548	0.01%	0.00%	9334.85	31548	0.01%	0.00%	5160.34	31548	0.01%	0.00%	3605.51
t4p14r2h4	42728	0.00%	0.00%	2744.60	42728	0.00%	0.00%	100.31	42728	0.00%	0.00%	129.75
t5p10r2h4	31790	19.02%	0.00%	21601.96	31790	19.02%	0.00%	3655.29	33417	22.96%	4.87%	3608.55
t5p11r8h3	38459	0.00%	0.00%	272.15	38459	0.00%	0.00%	38.29	38459	0.00%	0.00%	18.60
t5p12r2h2	25187	3.50%	0.00%	21601.20	25303	3.94%	0.46%	3603.13	25281	3.86%	0.37%	3603.98
t5p13r15h2	16698	0.00%	0.00%	8527.61	16698	0.00%	0.00%	3780.10	16698	0.00%	0.00%	3597.21
t5p14r2h1	17306	0.00%	0.00%	177.68	17306	0.00%	0.00%	73.35	17306	0.00%	0.00%	103.02
t6p10r6h2	21760	18.86%	0.00%	21600.98	21760	18.86%	0.00%	3621.75	21760	18.86%	0.00%	3603.11
t6p11r15h2	17924	4.11%	0.00%	21601.05	18908	9.10%	5.20%	4299.18	17924	4.11%	0.00%	3603.28
t6p12r10h2	23912	20.80%	0.00%	21601.10	23912	20.80%	0.00%	3725.56	23912	20.80%	0.00%	3602.84
t6p13r2h2	24401	26.53%	1.75%	21601.11	23975	25.22%	0.00%	3604.80	23975	25.22%	0.00%	3603.17
t6p14r2h1	23633	25.36%	0.00%	21600.98	34879	49.43%	32.24%	5962.40	23633	25.36%	0.00%	3603.09
Average	29016	6.92%	0.09%	11252.29	29612	8.33%	1.90%	3125.78	29081	7.07%	0.26%	2086.88

Table 3: Results 15 to 19 stopping points with Gurobi.

Instance	Model				Enumeration of routes				MILP-based heuristic			
	UB	Gap	Gap*	Time	UB	Gap	Gap*	Time	UB	Gap	Gap*	Time
t3p15r13h2	31927	0.00%	0.00%	24.61	31927	0.00%	0.00%	7.06	31927	0.00%	0.00%	3.33
t3p16r2h1	13579	0.01%	0.00%	7.50	13579	0.01%	0.00%	5.83	13740	1.18%	1.17%	5.44
t3p17r6h3	25404	0.00%	0.00%	28.59	25404	0.00%	0.00%	7.33	25404	0.00%	0.00%	4.92
t3p18r2h4	31463	0.01%	0.00%	38.92	31463	0.01%	0.00%	44.23	31463	0.01%	0.00%	22.73
t3p19r2h5	41170	0.00%	0.00%	68.39	41170	0.00%	0.00%	69.73	41170	0.00%	0.00%	46.42
t4p15r2h1	36256	3.40%	0.00%	21602.35	36440	3.89%	0.50%	3615.64	36314	3.55%	0.16%	3609.60
t4p16r2h2	30443	5.02%	0.00%	21601.56	30443	5.02%	0.00%	3607.26	30443	5.02%	0.00%	3605.56
t4p17r11h3	38010	0.00%	0.00%	35.03	38010	0.00%	0.00%	16.24	38010	0.00%	0.00%	12.44
t4p18r2h4	39382	0.00%	0.00%	2975.52	39382	0.00%	0.00%	94.70	39382	0.00%	0.00%	152.21
t4p19r15h1	5008	0.00%	0.00%	29.21	5008	0.00%	0.00%	45.79	5008	0.00%	0.00%	38.17
t5p15r2h2	24939	6.12%	0.00%	21600.88	24939	6.12%	0.00%	3603.64	24939	6.12%	0.00%	3602.84
t5p16r15h2	18440	3.89%	0.00%	21601.04	18440	3.89%	0.00%	3758.01	18440	3.89%	0.00%	3603.47
t5p17r6h1	22620	1.24%	0.00%	21600.97	22620	1.24%	0.00%	3622.81	22620	1.24%	0.00%	3603.14
t5p18r6h1	17834	0.00%	0.00%	8988.55	17834	0.00%	0.00%	3623.59	17834	0.00%	0.00%	3603.36
t5p19r16h1	26184	0.00%	0.00%	413.49	26184	0.00%	0.00%	1536.10	26184	0.00%	0.00%	1167.69
t6p15r7h1	29656	18.67%	0.00%	21601.23	29656	18.67%	0.00%	6684.10	29656	18.67%	0.00%	5651.25
t6p16r6h1	44038	59.12%	0.26%	21601.10	43922	59.01%	0.00%	13490.60	43922	59.01%	0.00%	13299.16
t6p17r10h1	14914	9.90%	0.00%	21601.26	14914	9.90%	0.00%	3702.08	15206	11.63%	1.92%	3605.14
t6p18r15h3	58217	0.00%	0.00%	17614.31	58217	0.00%	0.00%	1890.18	58217	0.00%	0.00%	567.87
t6p19r2h1	44639	41.95%	0.00%	21601.15	44639	41.95%	0.00%	7731.83	44639	41.95%	0.00%	8555.10
Average	29706	7.47%	0.01%	11231.78	29710	7.49%	0.03%	2857.84	29726	7.61%	0.16%	2737.99

Table 4: Results 20 to 24 stopping points with Gurobi.

Instance	Model				Enumeration of routes				MILP-based heuristic			
	UB	Gap	Gap*	Time	UB	Gap	Gap*	Time	UB	Gap	Gap*	Time
t3p20r16h2	29646	0.00%	0.00%	26.76	29646	0.00%	0.00%	54.90	29646	0.00%	0.00%	4.65
t3p21r13h5	46690	0.00%	0.00%	832.15	46690	0.00%	0.00%	54.13	46690	0.00%	0.00%	13.67
t3p22r6h1	17668	0.00%	0.00%	129.64	17668	0.00%	0.00%	139.97	17668	0.00%	0.00%	132.51
t3p23r12h4	39934	0.00%	0.00%	87.05	39934	0.00%	0.00%	923.88	39934	0.00%	0.00%	16.42
t3p24r7h5	49320	0.00%	0.00%	140.05	49320	0.00%	0.00%	91.47	49320	0.00%	0.00%	95.88
t4p20r8h1	22496	0.00%	0.00%	61.66	22496	0.00%	0.00%	35.84	22496	0.00%	0.00%	32.67
t4p21r8h2	25530	0.00%	0.00%	550.00	25530	0.00%	0.00%	101.32	25770	0.93%	0.93%	184.91
t4p22r2h1	47337	8.17%	0.00%	21601.75	47337	8.17%	0.00%	3608.51	47337	8.17%	0.00%	3606.67
t4p23r10h2	26361	0.00%	0.00%	2638.69	26361	0.00%	0.00%	197.95	26361	0.00%	0.00%	247.25
t4p24r2h1	16181	0.00%	0.00%	591.31	16181	0.00%	0.00%	211.23	16181	0.00%	0.00%	245.34
t5p20r2h1	27539	16.22%	0.00%	21601.58	29306	21.27%	6.03%	3604.26	27561	16.29%	0.08%	3605.68
t5p21r2h2	19450	0.00%	0.00%	10429.42	19450	0.00%	0.00%	3194.34	19450	0.00%	0.00%	3603.33
t5p22r13h2	29059	35.34%	0.00%	21601.17	29059	35.34%	0.00%	4732.20	29059	35.34%	0.00%	4156.55
t5p23r2h1	34691	0.00%	0.00%	3337.14	34691	0.00%	0.00%	356.48	34691	0.00%	0.00%	459.74
t5p24r2h2	33006	25.76%	0.00%	21601.19	33006	25.76%	0.00%	3605.08	33006	25.76%	0.00%	3604.01
t6p20r2h2	45893	63.34%	37.49%	21601.21	45893	63.34%	37.49%	4601.85	28686	41.35%	0.00%	3604.55
t6p21r2h1	8364	0.00%	0.00%	178.25	8364	0.00%	0.00%	313.04	8364	0.00%	0.00%	216.98
t6p22r11h2	49832	61.90%	0.33%	21601.44	49669	61.77%	0.00%	11443.31	51435	63.09%	3.43%	9546.47
t6p23r2h2	36146	56.35%	34.94%	21601.54	23516	32.91%	0.00%	3611.15	23516	32.91%	0.00%	3604.42
t6p24r16h1	39984	56.24%	32.39%	21601.57	43380	59.67%	37.68%	22171.90	27034	35.28%	0.00%	3605.12
Average	32256	16.17%	5.26%	9590.68	31875	15.41%	4.06%	3152.64	30210	12.96%	0.22%	2029.34

Table 5: Results for real instances with Gurobi.

Instance	Model				Enum				Heuristic			
	UB	Gap	Gap*	Time	UB	Gap	Gap*	Time	UB	Gap	Gap*	Time
t3p16r10h1	37180	0.00%	0.00%	3849.72	38080	2.36%	2.36%	21609.22	40360	7.88%	7.88%	109.15
t3p16r6h1	40360	0.00%	0.00%	150.80	40360	0.00%	0.00%	625.86	40360	0.00%	0.00%	101.49
t3p16r6h4	47056	0.00%	0.00%	2869.12	47056	0.00%	0.00%	701.86	47056	0.00%	0.00%	113.80
t3p16r6h6	47031	0.00%	0.00%	1933.36	47031	0.00%	0.00%	714.26	47031	0.00%	0.00%	118.77
t3p16r8h1	30860	0.00%	0.00%	6359.33	34480	10.50%	10.50%	21600.98	30860	0.00%	0.00%	5470.31
t4p16r10h1	26040	0.00%	0.00%	8598.96	26180	0.53%	0.53%	21600.02	26620	2.18%	2.18%	139.48
t4p16r6h1	26620	0.00%	0.00%	3398.16	30520	12.78%	12.78%	21601.42	26620	0.00%	0.00%	1822.59
t4p16r6h4	46816	17.43%	0.00%	21602.37	46816	17.43%	0.00%	21600.75	46816	17.43%	0.00%	214.08
t4p16r6h6	47031	8.68%	0.00%	21602.26	47031	8.68%	0.00%	21601.54	47031	8.68%	0.00%	2880.24
t4p16r8h1	36000	91.43%	0.00%	21602.24	—	—	—	21603.45	39340	92.16%	8.49%	21606.61
t5p16r10h1	28780	98.75%	7.51%	21602.49	—	—	—	21603.08	26620	98.65%	0.00%	3077.33
t5p16r6h1	26620	0.00%	0.00%	5544.74	—	—	—	21602.18	26620	0.00%	0.00%	3104.81
t5p16r6h4	46816	17.43%	0.00%	21602.51	—	—	—	21602.34	46816	17.43%	0.00%	2163.98
t5p16r6h6	47031	6.04%	0.00%	21601.95	47031	6.04%	0.00%	14548.33	47031	6.04%	0.00%	1042.07
t5p16r8h1	32580	87.97%	0.00%	21602.45	—	—	—	21604.11	34280	88.57%	4.96%	21852.29
Average	37788	21.85%	0.50%	12261.36	40458	5.83%	2.62%	16947.96	38231	22.60%	1.57%	4254.47

Table 6: Results 5 to 9 stopping points with HiGHS.

Instance	Model				Enum				Heuristic			
	UB	Gap	Gap*	Time	UB	Gap	Gap*	Time	UB	Gap	Gap*	Time
t2p5r2h1	31922	0.00%	0.00%	29.12	31922	0.00%	0.00%	16.03	31922	0.00%	0.00%	20.60
t2p6r7h3	42078	0.00%	0.00%	7.55	42078	0.00%	0.00%	13.49	42078	0.00%	0.00%	6.45
t2p7r8h1	17570	0.00%	0.00%	109.02	17570	0.00%	0.00%	245.18	17570	0.00%	0.00%	57.28
t2p8r2h2	20078	0.00%	0.00%	5.74	20078	0.00%	0.00%	4.31	20078	0.00%	0.00%	3.09
t2p9r2h3	37235	0.00%	0.00%	105.75	37235	0.00%	0.00%	17.43	37235	0.00%	0.00%	15.44
t3p5r7h1	7731	0.00%	0.00%	27.97	7731	0.00%	0.00%	49.24	7962	2.90%	2.90%	19.71
t3p6r2h1	23268	10.21%	0.00%	21601.63	23268	10.21%	0.00%	7900.97	23532	11.22%	1.12%	3797.13
t3p7r6h3	31411	0.00%	0.00%	160.32	31411	0.00%	0.00%	23.42	31411	0.00%	0.00%	24.85
t3p8r7h1	8474	0.00%	0.00%	157.29	8474	0.00%	0.00%	187.13	8474	0.00%	0.00%	83.44
t3p9r7h3	19386	0.00%	0.00%	24.48	19386	0.00%	0.00%	26.05	19386	0.00%	0.00%	21.97
t4p5r7h2	32670	48.33%	23.78%	21601.45	24902	32.21%	0.00%	21600.00	24902	32.21%	0.00%	3606.64
t4p6r7h1	—	—	—	21603.07	37599	99.97%	0.11%	23241.15	37556	99.97%	0.00%	21634.41
t4p7r7h4	36082	0.00%	0.00%	1106.64	36082	0.00%	0.00%	616.51	36082	0.00%	0.00%	51.83
t4p8r2h1	17313	6.83%	0.00%	21602.12	17313	6.83%	0.00%	6570.81	18001	10.39%	3.82%	3603.81
t4p9r2h1	9037	0.00%	0.00%	7446.35	9037	0.00%	0.00%	1157.28	9037	0.00%	0.00%	1319.13
t5p5r2h5	37452	0.00%	0.00%	15120.32	37452	0.00%	0.00%	1859.17	37452	0.00%	0.00%	3602.85
t5p6r6h1	—	—	—	21602.48	19450	31.88%	0.00%	23999.81	20406	35.07%	4.68%	21629.62
t5p7r2h2	—	—	—	21603.03	29875	44.97%	0.00%	14416.64	33475	50.89%	10.75%	14422.00
t5p8r8h3	17469	10.10%	0.00%	21601.13	17469	10.10%	0.00%	4485.57	17469	10.10%	0.00%	3607.34
t5p9r2h3	15935	8.91%	0.00%	21602.58	15935	8.91%	0.00%	3623.61	15935	8.91%	0.00%	3613.84
Average	23830	4.96%	1.40%	9855.90	24213	12.25%	0.01%	5502.69	24498	13.08%	1.16%	4057.07

Table 7: Results 10 to 14 stopping points with HiGHS

Instance	Model				Enum				Heuristic			
	UB	Gap	Gap*	Time	UB	Gap	Gap*	Time	UB	Gap	Gap*	Time
t3p10r6h3	37612	0.00%	0.00%	215.15	37612	0.00%	0.00%	36.85	37612	0.00%	0.00%	36.34
t3p11r2h1	38040	7.26%	0.00%	21601.89	38040	7.26%	0.00%	3613.08	38040	7.26%	0.00%	3607.98
t3p12r2h3	36985	0.00%	0.00%	37.19	36985	0.00%	0.00%	8.94	36985	0.00%	0.00%	6.34
t3p13r12h2	17968	0.00%	0.00%	803.36	17968	0.00%	0.00%	180.95	17968	0.00%	0.00%	131.26
t3p14r8h5	39947	60.50%	20.91%	21601.47	31596	50.06%	0.00%	21600.85	32104	50.85%	1.58%	679.61
t4p10r6h3	27620	0.00%	0.00%	18923.66	27620	0.00%	0.00%	468.79	27620	0.00%	0.00%	106.36
t4p11r2h2	40927	22.70%	0.00%	21602.68	40927	22.70%	0.00%	3677.26	40927	22.70%	0.00%	3631.12
t4p12r2h2	34230	0.00%	0.00%	4144.65	34230	0.00%	0.00%	3611.27	34230	0.00%	0.00%	3607.20
t4p13r8h1	41148	32.67%	23.12%	21601.85	31636	12.43%	0.00%	5760.04	31744	12.72%	0.34%	3607.11
t4p14r2h4	45277	11.51%	5.63%	21602.29	42728	6.23%	0.00%	3744.28	42728	6.23%	0.00%	3559.94
t5p10r2h4	—	—	—	21603.84	35259	44.38%	9.18%	4275.50	32023	38.76%	0.00%	3661.23
t5p11r8h3	38459	0.00%	0.00%	1309.84	38459	0.00%	0.00%	362.57	38459	0.00%	0.00%	204.43
t5p12r2h2	25187	4.22%	0.00%	21600.99	26004	7.23%	3.14%	3605.61	25458	5.24%	1.06%	3606.37
t5p13r15h2	16698	0.01%	0.00%	18697.35	16698	0.01%	0.00%	3893.26	16698	0.01%	0.00%	3603.79
t5p14r2h1	17306	0.00%	0.00%	800.36	17306	0.00%	0.00%	957.93	17306	0.00%	0.00%	1105.21
t6p10r6h2	21760	22.54%	0.00%	21600.56	21760	22.54%	0.00%	3649.16	21760	22.54%	0.00%	3603.74
t6p11r15h2	18856	24.19%	0.00%	21601.33	18908	24.40%	0.28%	4628.37	18908	24.40%	0.28%	3604.71
t6p12r10h2	30773	45.47%	22.30%	21601.75	23912	29.82%	0.00%	3828.01	23912	29.82%	0.00%	3604.80
t6p13r2h2	26000	0.00%	0.00%	9725.83	26210	0.80%	0.80%	3609.13	26501	1.89%	1.89%	3604.71
t6p14r2h1	23633	29.32%	0.00%	21601.30	25939	35.60%	8.89%	3606.52	23633	29.32%	0.00%	3603.94
Average	30443	13.70%	3.79%	14613.87	29490	13.17%	1.11%	3755.92	29231	12.59%	0.26%	2458.81

Table 8: Results 15 to 19 stopping points with HiGHS.

Instance	Model				Enum				Heuristic			
	UB	Gap	Gap*	Time	UB	Gap	Gap*	Time	UB	Gap	Gap*	Time
t3p15r13h2	31927	0.00%	0.00%	193.38	31927	0.00%	0.00%	26.45	31927	0.00%	0.00%	13.42
t3p16r2h1	13579	0.01%	0.00%	13.26	13579	0.01%	0.00%	39.40	13740	1.18%	1.17%	23.80
t3p17r6h3	25404	0.00%	0.00%	272.17	25404	0.00%	0.00%	39.49	25404	0.00%	0.00%	31.65
t3p18r2h4	31463	0.01%	0.00%	11337.83	31463	0.01%	0.00%	348.16	31463	0.01%	0.00%	236.50
t3p19r2h5	41170	0.00%	0.00%	2213.03	41170	0.00%	0.00%	324.19	41170	0.00%	0.00%	272.75
t4p15r2h1	43584	5.08%	0.00%	21603.62	—	—	—	14422.03	—	—	—	10820.41
t4p16r2h2	31788	14.68%	3.66%	21602.57	30626	11.44%	0.00%	3615.43	30776	11.87%	0.49%	3609.96
t4p17r11h3	38010	0.00%	0.00%	988.11	38010	0.00%	0.00%	443.72	38010	0.00%	0.00%	213.38
t4p18r2h4	53286	35.74%	26.09%	21602.72	39382	13.05%	0.00%	1316.77	39382	13.05%	0.00%	1095.58
t4p19r15h1	5008	0.00%	0.00%	471.64	5008	0.00%	0.00%	104.94	5008	0.00%	0.00%	70.13
t5p15r2h2	25629	41.76%	2.69%	21602.12	24939	40.15%	0.00%	3606.17	24939	40.15%	0.00%	3603.40
t5p16r15h2	18440	14.12%	0.00%	21601.80	18440	14.12%	0.00%	3937.82	18440	14.12%	0.00%	3604.47
t5p17r6h1	22620	20.52%	0.00%	21601.29	22620	20.52%	0.00%	3646.49	22620	20.52%	0.00%	3603.91
t5p18r6h1	17834	8.59%	0.00%	21601.15	17834	8.59%	0.00%	3649.61	17834	8.59%	0.00%	3604.03
t5p19r16h1	26184	46.04%	0.00%	21601.12	26184	46.04%	0.00%	11312.82	26184	46.04%	0.00%	3341.91
t6p15r7h1	29656	49.51%	0.00%	21601.32	29656	49.51%	0.00%	15937.69	29656	49.51%	0.00%	11668.72
t6p16r6h1	43922	47.32%	0.00%	21601.39	43922	47.32%	0.00%	18831.17	44316	47.79%	0.89%	11713.37
t6p17r10h1	25401	51.97%	40.21%	21601.54	15186	19.66%	0.00%	3747.59	15788	22.73%	3.81%	3605.61
t6p18r15h3	58217	34.69%	0.00%	21601.57	58217	34.69%	0.00%	9691.64	58217	34.69%	0.00%	6930.11
t6p19r2h1	46872	18.91%	4.76%	21601.83	44639	14.85%	0.00%	14406.93	47233	19.53%	5.49%	10807.01
Average	31500	19.45%	3.87%	14815.67	29379	16.84%	0.00%	5472.43	29585	17.36%	0.62%	3943.51

Table 9: Results 20 to 24 stopping points with HiGHS.

Instance	Model				Enum				Heuristic			
	UB	Gap	Gap*	Time	UB	Gap	Gap*	Time	UB	Gap	Gap*	Time
t3p20r16h2	29646	0.00%	0.00%	332.85	29646	0.00%	0.00%	116.24	29646	0.00%	0.00%	30.23
t3p21r13h5	46690	0.00%	0.00%	786.05	46690	0.00%	0.00%	284.16	46690	0.00%	0.00%	112.31
t3p22r6h1	17668	0.00%	0.00%	2278.73	17668	0.00%	0.00%	348.71	17668	0.00%	0.00%	655.95
t3p23r12h4	39934	0.00%	0.00%	4014.93	39934	0.00%	0.00%	2844.10	39934	0.00%	0.00%	242.97
t3p24r7h5	49320	0.00%	0.00%	4498.61	49320	0.00%	0.00%	368.23	49320	0.00%	0.00%	55.08
t4p20r8h1	22496	0.00%	0.00%	179.85	22496	0.00%	0.00%	585.00	22496	0.00%	0.00%	420.56
t4p21r8h2	25530	0.01%	0.00%	2050.53	25530	0.01%	0.00%	357.02	25770	0.94%	0.93%	267.76
t4p22r2h1	—	—	—	21602.63	49984	26.26%	0.00%	14412.41	54460	32.32%	8.22%	10812.40
t4p23r10h2	26361	0.00%	0.00%	6939.09	26361	0.00%	0.00%	837.77	26361	0.00%	0.00%	512.38
t4p24r2h1	16181	0.01%	0.00%	1491.19	16181	0.01%	0.00%	612.33	16181	0.01%	0.00%	794.18
t5p20r2h1	27552	20.15%	0.00%	21602.15	27601	20.29%	0.18%	3607.70	27874	21.07%	1.16%	3608.59
t5p21r2h2	41250	58.80%	52.85%	21602.31	19450	12.62%	0.00%	3609.05	19450	12.62%	0.00%	3605.09
t5p22r13h2	29059	0.00%	0.00%	8105.17	29059	0.00%	0.00%	8856.30	29059	0.00%	0.00%	4798.11
t5p23r2h1	34691	1.06%	0.00%	21601.47	34691	1.06%	0.00%	1462.41	34691	1.06%	0.00%	1874.56
t5p24r2h2	47086	58.72%	29.90%	21603.01	33006	41.11%	0.00%	3610.27	33006	41.11%	0.00%	3605.40
t6p20r2h2	45893	64.12%	37.66%	21602.02	28609	42.44%	0.00%	3612.75	28677	42.58%	0.24%	3606.67
t6p21r2h1	8364	0.00%	0.00%	428.79	8364	0.00%	0.00%	372.86	8364	0.00%	0.00%	463.88
t6p22r2h1	30913	8.72%	0.00%	21601.99	39404	28.39%	21.55%	13268.94	30913	8.72%	0.00%	14285.27
t6p23r2h2	36146	59.28%	41.17%	21602.55	21961	32.98%	3.17%	3615.90	21265	30.78%	0.00%	3607.86
t6p24r16h1	—	—	—	21602.21	28784	40.43%	0.00%	4828.06	30442	43.67%	5.45%	3607.89
Average	31932	15.05%	8.98%	11276.31	29737	12.28%	1.24%	3380.51	29613	11.90%	0.80%	2848.36

Table 10: Results for real instances with HiGHS.

Instance	Model				Enum				Heuristic			
	UB	Gap	Gap*	Time	UB	Gap	Gap*	Time	UB	Gap	Gap*	Time
t3p16r10h1	48220	83.91%	21.03%	21602.35	38080	79.63%	0.00%	21614.10	40360	80.78%	5.65%	5397.84
t3p16r6h1	40620	0.79%	0.64%	21601.88	40360	0.15%	0.00%	4112.36	40360	0.15%	0.00%	5277.42
t3p16r6h4	47056	4.54%	0.00%	21601.89	47056	4.54%	0.00%	2807.73	47056	4.54%	0.00%	1332.36
t3p16r6h6	47031	0.00%	0.00%	15116.97	47031	0.00%	0.00%	1623.92	47031	0.00%	0.00%	381.78
t3p16r8h1	—	—	—	21601.97	47140	99.61%	12.60%	21676.41	41200	99.56%	0.00%	21788.41
t4p16r10h1	—	—	—	21602.64	26180	34.99%	0.00%	21600.02	26620	36.06%	1.65%	6502.87
t4p16r6h1	51320	96.18%	40.53%	21603.26	30520	93.58%	0.00%	21601.35	42180	95.35%	27.64%	21683.16
t4p16r6h4	50716	85.71%	7.69%	21603.13	46816	84.52%	0.00%	21600.67	46816	84.52%	0.00%	21649.13
t4p16r6h6	47031	86.30%	0.00%	21603.45	47031	86.30%	0.00%	21603.56	47031	86.30%	0.00%	22991.51
t4p16r8h1	—	—	—	21602.91	—	—	—	21602.10	41660	99.98%	0.00%	21669.59
t5p16r10h1	—	—	—	21603.49	—	—	—	21600.16	32540	—	0.00%	23855.96
t5p16r6h1	46820	91.63%	30.50%	21603.50	—	—	—	21602.24	32540	87.96%	0.00%	23699.86
t5p16r6h4	47116	83.59%	0.64%	21603.58	—	—	—	21601.03	46816	83.48%	0.00%	8770.77
t5p16r6h6	47031	81.87%	0.00%	21602.72	47031	81.87%	0.00%	21601.85	47031	81.87%	0.00%	4966.51
t5p16r8h1	—	—	—	21603.92	—	—	—	21603.47	34820	99.88%	0.00%	22889.95
Average	47296	61.45%	10.10%	21170.51	41724	56.52%	1.26%	17856.73	40937	67.17%	2.33%	14190.47